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ECON412 International Economics: Trade

Trimester 2, 2018

Lecture Week #5: The Eaton-Kortum Model

21 August 2018

Overview

- **Readings:** Feenstra Chapter 6, pp 172-173. Eaton and Kortum (2012).
- Extend the Ricardian model to many goods and *many countries*.
- Derive the gravity equation.

1. Introduction

- The Ricardian model with many goods and two countries, as in DornBusch, Fischer, and Samuelson (1977)
  - has a nice feature: there is a borderline point z' on the chain of comparative advantage $A(z) = \frac{a^*(z)}{a(z)}$.
- Home produces goods $z \in [0, z']$ and Foreign produces goods $z^* \in [z', 1]$, when we rank the relative productivity such that $A(z)$ declines as z increases.
- The chain of comparative advantage no longer works with many goods and many countries.
- That is, pair-wise ranking does not uniquely determine patterns of trade if there are more than 2 countries.

Example

Consider labor requirements in 3 countries for 3 goods

	Canada	**U.S.**	**Mexico**
Baseball	10	10	10
Footbal	5	7	3
Soccer ball	4	3	2

- Pair-wise comparative advantage suggests multiple equilibrium:
  - An equilibrium: Mexico makes Soccer ball, U.S. makes Baseball, Canada makes Football.
  - Another equilibrium: Mexico makes Football, U.S. makes Soccer ball, Canada makes Baseball.

- Need a way to model trade flows for all countries. That is Eaton and Kortum (2002)'s probabilistic approach.
- Their model led to "Ricardian Revival", it is an important contribution to the theory of comparative advantage.
- Key idea: take the $a(z)$'s as the realizations of random variables drawn from certain probability distribution.
    - No longer need to keep track of all $a(z)$ (z represents infinitely many).

- Key questions to answer from the EK model
  - quantifying the gains from trade.
  - how changes in trade barriers affect these gains.
  - how technological change in trading partners affect these gains.

2. The Eaton-Kortum (2002) model

- We present a simplified version of the original model, as in Eaton-Kortum (2012, JEP).
- Same assumptions as in 2-country-many-good case, but now allow for many countries.

2-1. Geography

- N countries, indexed as $i = 1, \cdots, N$. Index a country with n if it is a destination.
- A continuum of goods, $j \in [0, 1]$.
- Iceberg trade cost: delivering one unit of a good from country i to country n requires producing d_{ni} units in country i. Assume
  - triangle inequality: $d_{ni} < d_{nk} d_{ki}$. In addition, $d_{ii} = 1$.
  - If d_{ni} increases with the distance between i and n, it is called the "gravity".
- Perfect competition in all markets.

2-2. Technology

- Producing one unit of good j in country i requires $a_i(j)$ units of labor. Let w_i be the wage of a worker in country i.
- We treat $a_i(j)$ as a random variable with probability $Pr[a_i(j) \leq x] = F_i(x)$.
  - We do not specify which goods country i is good at, only the probability that the country is good at producing a good.
  - By the law of large numbers, this probability is the fraction of goods that require less than x units of labor to produce in country i.

Which distribution should be used?

- Want the distribution of the price of a good to stay within the same distribution family as $a(z)$.
- That price is the minimum of costs of acquiring the good across all countries.
- Also want a tractable expression for the probability that a particular country is the lowest-cost source.
- This led EK to choose the *extreme value* distributions, $F(x) = 1 - e^{-x^{\theta}}$.
- Extreme value distributions are "closed" under maximization and minimization.
  - The maximum (or minimum) of several random variables are also extrem-value distributed.

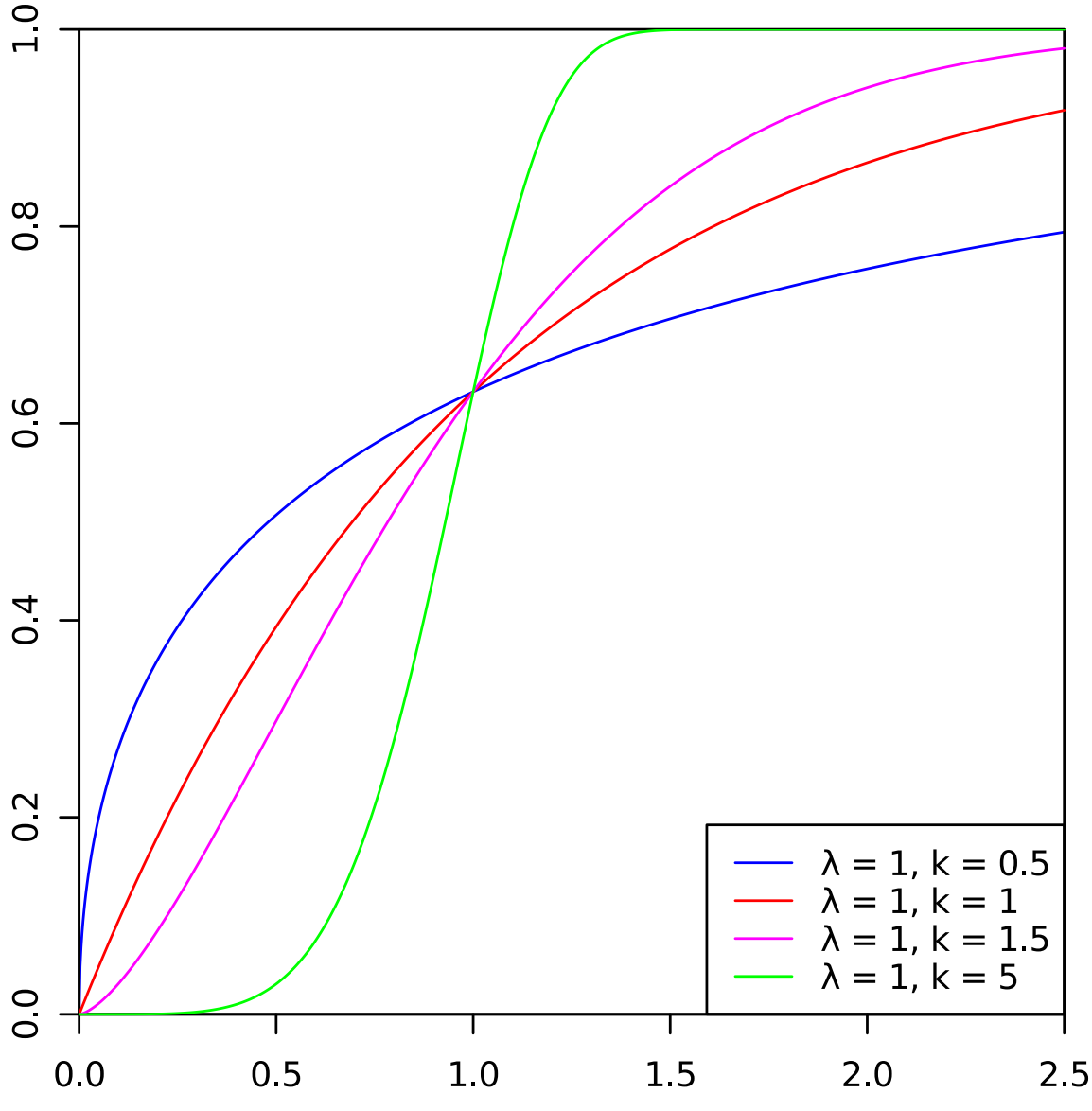
- Productivity follows the Fréchet distribution, then unit labor requirement (inverse productivity) follows the Weibull distribution, with cumulative distribution function

$$Pr[a_i(j) \leq x] = 1 - e^{-(A_i x)^\theta}, \text{ with } A_i > 0, \theta > 1.$$

- The probability density function of labor requirement is $f(x) = \theta A_i^\theta x^{\theta-1} e^{-(A_i x)^\theta}$.
  - Mean: $\frac{1}{A_i}\Gamma(1+\frac{1}{\theta})$. Variance: $\frac{1}{A_i^2}\Gamma(1+\frac{2}{\theta}) - [\Gamma(1+\frac{1}{\theta})]^2$. Median: $\frac{1}{A_i}[\ln(2)]^\theta$.
  - Note: Gamma function is $\Gamma(s) = \int_0^\infty x^{s-1} e^{-x} dx$.

- *Absolute advantage*: scale parameter A_i governs the location of distribution, a larger A_i implies that labor requirement of producing any good j is likely lower.
    - This captures the fact: some countries are overall more productive than others in producing a large number of goods.
- *Comparative advantage*: Shape parameter θ, common to all countries, reflects (inversely) the dispersion of labor requirement in producing different goods.
    - A higher value of θ implies that a country's labor requirements are closer to the mean, weakening the force of comparative advantage.

- Weibull distribution, CDF is $F(x) = 1 - e^{-(x/\lambda)^k}$.



2-3. Preferences

- Cobb-Douglas preference, $U = \int_0^1 \ln[q(j)]dj$.
- This implies that consumers spend an equal amount of their income on all goods.

2-4. Prices

- Market is perfectly competitive. Serving one unit of good j produced in country i to country n costs $c_{ni}(j) = d_{ni}a_i(j)w_i$.
  - In country i, wage and trade costs are the same across goods.
- $p_{ni}(j) = c_{ni}(j)$ is the unit price that country n pays if it buys good j from country i.

- Price $p_{ni}(j)$ has a similar distribution to $a_i(j)$

$$G_{ni}(p) = Pr[p_{ni}(j) \leq p] = Pr[a_i(j) \leq \frac{p}{d_{ni}w_i}] = 1 - e^{-(A_{ni}p)^{\theta}},$$

$$\text{with } A_{ni} = \frac{A_i}{w_i d_{ni}}.$$

- Country i charges different prices for the same good j, depending on the destination.

- The price that country n pays for good j is $p_n(j) = \min\{p_{ni}(j), i = 1, \cdots, N\}$.

- The distribution of $p_n(j)$ is

$$G_n(p) = Pr[p_n(j) \leq p] = 1 - Pr[p_{n1} \geq p, \cdots, p_{nN} \geq p]$$
$$= 1 - \prod_{i=1}^{N}[1 - G_{ni}(p)].$$

Or, $G_n(p) = 1 - e^{-\Phi_n p^\theta}$, where $\Phi_n = \sum_{i=1}^{N} A_{ni}^\theta = \sum_{i=1}^{N} \left(\frac{A_i}{w_i d_{ni}}\right)^\theta$.

- Price distribution in country n is also Weibull, with scale parameter $\Phi_n^{1/\theta}$.
- Parameter Φ_n summarizes how each of the following factors governs price $p_n(j)$
  - A_i, state of the technology around the world;
  - w_i, input costs around the world;
  - d_{ni}, geographic barriers of trade.
- Higher A_i, lower w_i, and smaller d_{ni}, all translate into a lower $p_n(j)$.

Trade flows and aggregate price

We skip derivations.

1) The probability that country i is the lowest-cost source of a good in country n is $Pr[p_{ni}(j) \leq \min\{p_{nk}(j), k \neq i\}]$, which equals

$$\pi_{ni} = \frac{A_{ni}^{\theta}}{\Phi_n} = \frac{A_{ni}^{\theta}}{\sum_{k=1}^{N} A_{nk}^{\theta}}.$$

- By the law of large numbers, π_{ni} is also the fraction of goods that n imports from i.
- With a unit continuum of goods and (symmetric) Cobb-Douglas preference, π_{ni} is the *trade share*, i.e., country n's expenditure on goods imported from i as a fraction of country n's total spending.

- Country n's income is $Y_n = w_n L_n$. Let X_{ni} be the value of trade from i to n, then $X_{ni} = \pi_{ni} w_n L_n$. That is
$$X_{ni} = \frac{A_{ni}^{\theta}}{\sum_{k=1}^{N} A_{nk}^{\theta}} w_n L_n.$$
- X_{ni} is the trade flow (value) from i to n, a function of geography, productivities and wages.

2) The exact price index under the Cobb-Douglas utility function is

$$p_n = \exp\left(\int_0^\infty \ln(p) dG_n(p)\right) = \frac{\gamma}{\Phi_n^{1/\theta}}, \text{ with } \gamma = e^{-\varepsilon/\theta},$$

$$\varepsilon \text{ is the Euler's number } (= 0.5772\cdots).$$

- This is the overall price level in country n.

2-3. Equilibrium

- Wages and bilateral trade flows are endogenous. Other variables are exogenous.
- Income in country i is, $Y_i = w_i L_i = \sum_{n=1}^{N} X_{ni}$. This becomes

$$w_i L_i = \sum_{n=1}^{N} \frac{A_{ni}^{\theta}}{\sum_{k=1}^{N} A_{nk}^{\theta}} w_n L_n.$$

- Note that $A_{ni} = \frac{A_i}{w_i d_{ni}}$.
- This is a system of N equations, solving them we obtain the wages. Need to solve it on computer.
- Once wages are solved for, trade flows then follow.

- With the above Cobb-Douglas utility function, indirect utility in country i is $U_i = w_i/p_i$, where p_i is aggregate price.
- For country i, we have $\pi_{ii} = \frac{A_{ii}^{\theta}}{\Phi_i}$ and $p_i = \frac{\gamma}{\Phi_i^{1/\theta}}$. From these two equations, we obtain the real wage in country i as
$$\frac{w_i}{p_i} = \gamma^{-1} A_i \pi_{ii}^{-\frac{1}{\theta}}.$$
  - A_i captures labor productivity.
  - $\pi_{ii}^{-1/\theta}$ measures the **gains from trade**. In autarky, $\pi_{ii} = 1$, then $\frac{w_i}{p_i}|_{\text{autarky}} = \gamma^{-1} A_i$.
  - The smaller is θ, the bigger the difference in technology across countries, hence the larger the gains.
  - $\theta = 4$, estimated based on Simonovska and Waugh (2011).

2-4. Special case: $d_{ni} = 1$.

- Suppose that there is no trade barrier, $d_{ni} = 1$ for all n and i.
- The relative wage between countries k and i can be shown to be given by

$$\frac{w_k}{w_i} = \left(\frac{A_k^\theta / L_k}{A_i^\theta / L_i} \right)^{\frac{1}{1+\theta}}.$$

- It can also be shown that the real wage is

$$\frac{w_i}{p_i} = \gamma^{-1} A_i^{\frac{\theta}{1+\theta}} \left[\sum_{k=1}^{N} \left(\frac{A_k L_k}{L_i} \right)^{\frac{\theta}{1+\theta}} \right]^{1/\theta}.$$

- The gains from trade again are determined by the relative labor productivity, relative labor endowment, and the elasticity parameter θ.

3. Quantitative analysis

- Questions the model can answer
  - How much do countries gain from trade, and how have these gains changed over time?
  - How would the gains from trade change, if trading barriers are eliminated?
  - To what extent do countries benefit from technological improvements of their trading partners?
  - (There are more)
- In order to quantify the gains from trade, the original model is needed, which features
  - preference with constant elasticity of substitution (CES).
  - Cobb-Douglas production function, The efficiency of production, TFP, is a random variable.
  - intermediate inputs in production.
  - traded sector (manufacture) and non-traded sector (services).

3-1. Gains from trade

- Share of manufactures is set to $\alpha = 0.2$ of the final spending.
- Share of manufactures labor cost in total is $\beta = 0.3$, the rest is largely intermediate inputs.
- $\theta = 4$.
- Home share, π_{ii}, measured as: gross manufacturing production less gross exports, divided by gross manufacturing production less net exports. In other words, spending on domestic manufactures out of total spending on manufactures.
- In EK's original model, measure of gains from trade is $\pi_{ii}^{-\alpha/(\beta\theta)}$.
- Gains from trade (relative to autarky) in data is measured as $100[(\pi_{ii}^{t})^{-1/6} - 1]$.
- Table 2 of Eaton-Kortum (2012). Gains arise from final use and manufactured goods as intermediate inputs.

Table 2

The Home Share of Spending on Manufactures and Gains from Trade

Country	World GDP share (%) in 2006	Home share of spending		Implied gains from trade	
		Level in 2006 (%)	Change since 1996 (percentage points)	Level in 2006 (%)	Change since 1996 (percentage points)
Austria	0.66	31.4	–16.2	21.3	8.1
Canada	2.60	49.1	–1.5	12.6	0.6
Czech Republic	0.29	42.6	–14.7	15.3	5.5
Denmark	0.56	25.6	–18.1	25.5	10.7
Estonia	0.03	2.5	–19.6	85.4	56.7
Finland	0.42	58.2	–7.3	9.4	2.1
France	4.60	56.9	–10.3	9.9	3.0
Germany	5.94	53.7	–16.4	10.9	4.8
Greece	0.54	52.7	–11.6	11.3	3.6
Hungary	0.23	26.0	–34.5	25.1	16.4
Iceland	0.03	27.9	–10.0	23.7	6.2
Ireland	0.46	39.6	9.9	16.7	–5.7
Italy	3.80	68.9	–7.1	6.4	1.7
Japan	8.88	84.9	–5.6	2.8	1.1
Korea	1.94	77.2	–0.7	4.4	0.1
Mexico	1.94	58.3	–7.9	9.4	2.3
New Zealand	0.22	53.6	–8.2	11.0	2.6
Norway	0.68	51.9	–2.5	11.6	0.9
Poland	0.69	53.4	–15.8	11.0	4.7
Portugal	0.41	50.8	–10.2	12.0	3.4
Slovenia	0.08	27.2	–15.5	24.3	9.0
Spain	2.51	62.8	–10.2	8.1	2.7
Sweden	0.81	49.2	–10.0	12.5	3.4
Switzerland	0.80	35.3	–20.0	18.9	8.6
United States	27.26	73.5	–8.3	5.3	1.9
All others	33.62				

3-2. Benefits of technological improvements

- To what extent an improvement in technology in one country can benefit others through trade?
- Counterfactual experiment: suppose that the United States becomes 10 percent more productive, e.g., $A_{US} = 1.1$. The world economy responds in two ways
  - The U.S. wage rises by about 30 percent relative to other countries' wages.
  - The U.S. real wage rises by about 6 percent, while real wages in other countries rise by only a small amount.
- The gains largely depends on the geography. Canada and Mexico will experience a real wage gain of 1/10 that in the U.S.
- Overall, the increase in the U.S. technology raises the GDP-weighted real wage around the world by 1.6 percent, with 8 percent of this gain experienced outside the U.S.

4. The Gravity Equation

- We already had a gravity equation from the EK model.
- With a unit continuum of goods $j \in [0.1]$ and symmetric Cobb-Douglas preference, π_{ni} becomes the trade share, i.e., country i's exports to country n as a fraction of country n's total spending. Let X_{ni} be the exports of country i to country n, and X_n be the total spending of country n. Recall that $A_{ni} = \frac{A_i}{w_i d_{ni}}$. Then

$$\frac{X_{ni}}{X_n} = \frac{A_{ni}^{\theta}}{\Phi_n} = \frac{A_{ni}^{\theta}}{\sum_{k=1}^{N} A_{nk}^{\theta}} = \frac{A_i^{\theta}(w_i d_{ni})^{-\theta}}{\sum_{k=1}^{N} A_k^{\theta}(w_k d_{nk})^{-\theta}}.$$

- The exporter's total sales is
$$Q_i = \sum_{m=1}^{N} X_{mi} = A_i^{\theta} w_i^{-\theta} \sum_{m=1}^{N} \frac{d_{mi}^{-\theta} X_m}{\Phi_m}.$$
Solve for $A_i^{\theta} w_i^{-\theta}$ in the above equation, then it plug into the trade share expression, also use $p_n = \gamma / \Phi_n^{1/\theta}$, we have
$$X_{ni} = \frac{(d_{ni}/p_n)^{-\theta} X_n Q_i}{\sum_{m=1}^{N} (d_{mi}/p_m)^{-\theta} X_m}.$$

- This is a variation of the **Gravity Equation**, with bilateral resistance d_{ni}, and multilateral resistance terms p_n (inward) and $\sum_{m=1}^{N} (d_{mi}/p_m)^{-\theta} X_m$ (outward).

- The gravity equation can be estimated. Variables are on the right-hand side are observables.

  - The trade barrier d_{ni} includes geographical distance, dummy variables indicating whether sharing borderlines, whether using the same language, whether in free-trade area, etc.
  - The interest is θ, the elasticity parameter.